Ziheng He

Phone: +86 199 3540 7446 | Email: heziheng261@mails.ucas.ac.cn
Research Internship - World Models / Video Generation / Embodied AI
Website: zihenghe04.github.io | **GitHub:** github.com/zihenghe04

Education

UCAS / Institute of Automation, CAS / NLPR | Ph.D. in Artificial Intelligence 2026 - Present

- **Research interests:** World models, Vision-Language-Action (VLA), and Multimodal Large Language Models (MLLMs).

Southeast University | B.Eng. in Software Engineering 2022.09 - 2026.06

- **Rank and honors:** 9/86 (Top 10%); Outstanding Graduate, Merit Student, and Outstanding Thesis.

Research Experience and Selected Work

GigaAI | Research Intern 2025.09 - Present

GigaBrain-0.7 Embodied Foundation Model | Co-author | arXiv 2608.15875

Three-system architecture trained on 37,256 hours across 16 robot types; System 3 supplies predicted visual states and task progress to the action policy.

- **World-model adaptation:** Trained **System 3 (5B World Value Model)** by continuing GigaWorld-1 on real and simulated robot videos for robot-centric prediction; extended the MoT value pathway with **trajectory-completion labels** for task progress.
- **Conditioning interface:** Aligned a last-frame visual subgoal with the downstream action chunk and paired it with a 1-bit progress signal, forming a **low-bandwidth interface** to inject world-model knowledge into System 1 without deployment-time rollout.
- **Post-training:** Applied **independent condition dropout** to both signals across unconditional, single-, and dual-condition modes; froze System 3 and updated only Systems 1/2 to limit dependence on auxiliary inputs.
- **Ablations:** Gift-wrapping success for Base / +Subgoal / +Value / Both was **0% / 20% / 60% / 80%**. With garment-folding success saturated at 100%, both conditions still improved task score **68.3% -> 88.3%** and time **107s -> 75s**.

arxiv.org/abs/2608.15875

FiveAges | Research Intern 2025.09 - Present

SKIP: Sparse Keyframe Interpolation for Efficient Embodied World Models | First Author | arXiv 2606.00664

Addresses compute cost and long-horizon drift in recursive dense rollouts with an event-preserving sparse-to-dense design.

- **Keyframe selection:** Fused **optical flow, DINOv3 semantics, and robot proprioception**; used KTS temporal segmentation and gripper-event protection to select keyframes, achieving **GripCov 0.999**.
- **Sparse-to-dense generation:** Generated only sparse key events with a video diffusion model, estimated inter-keyframe gaps with a gap predictor, and reconstructed dense rollouts through **action-conditioned interpolation**.
- **Results:** Achieved **4.16x** speedup and reduced FVD from **4177 to 458** against a recursively chunked dense Wan 2.2 baseline. Policies trained entirely on generated demonstrations lost only **1.3 / 6.7 pp** success in simulation / on real robots.

arxiv.org/abs/2606.00664 · zihenghe04.github.io

FiveagesWAM: Real-Time Interactive World Model

Built for online interaction, balancing fast response to new conditions with historical context for continuous prediction.

- **Interaction and memory:** Accepted streaming conditioning inputs and updated predictions across continuous chunks; combined **keyframe and sparse-frame inference**, injecting historical images as **memory** to preserve cross-chunk context.
- **Real-time performance:** Generated **8-frame chunks covering 0.8 s of video with 426 ms HTTP raw P95 end-to-end latency**, reaching **1.88x real time (faster than real time)**.

Huawei Nanjing Research Center, Celia Team | AI Algorithm Engineer Intern 2025.06 - 2025.09

On-Device LLM Inference Acceleration | EAGLE-3 / Fast-dLLM / CUDA

Optimized decoding latency, cache efficiency, and batch throughput for an on-device assistant under constrained compute.

- **Speculative decoding:** Reproduced EAGLE-3 with a lightweight draft model and multi-layer feature fusion, improving draft acceptance by about **8%**.
- **Parallel and kernel optimization:** Optimized KV cache and parallel decoding on Fast-dLLM and profiled CUDA memory access and kernels with Nsight; raised KV-cache hit rate from **70% to 85%** and batch throughput by about **17%**.
- **System validation:** Reduced average latency from **240 ms to 187 ms** in A/B tests on multi-turn Celia queries; the work entered the team's inference framework and was nominated for a technical innovation award.

Open Source and Public Projects

github.com/iOfficeAI/AionUi Project total: 32.3k Stars | 3.3k Forks

- **Contribution:** Authored the merged headless-Linux deployment guide for a 24/7 Cowork app supporting 20+ CLI agents, covering Xvfb, services, ngrok / SSH access, and bilingual docs.

github.com/zihenghe04/CCDash Project total: 71 Stars | 5 Forks

- **Built and maintained:** Created a unified usage dashboard for Claude Code, claude.ai, and API traffic, aggregating tokens, quotas, and cost; v0.9.2 added session status, endpoint detection, and macOS launch-on-login.